

**22      Ans: C**

When switch is open:

$$V = E - I r$$

$$1.5 = E - (0)r$$

$$E = 1.5 \text{ V}$$

When switch is closed:

$$V = E - I r$$

$$0.75 = 1.5 - (0.75/4.0)r$$

$$r = 4.0 \, \Omega$$

**23      Ans: B**